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Experiment Title : Demonstration of Switch Hardware and Setting up a Basic Local Area Network (LAN)

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Experiment No. 06: Demonstration of Switch Hardware and Setting up a Basic Local Area Network (LAN)

Aim

To identify the hardware interfaces of a Network Switch and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology in Cisco Packet Tracer.

Theory

What is a network switch?

A network switch is a networking device used to connect multiple devices within a Local Area Network (LAN). It operates mainly at the Data Link Layer (Layer 2) of the OSI model. Unlike a hub, a switch forwards data only to the intended destination device. This makes communication more efficient and reduces network congestion.

What are the main functions of a network switch?

The main function of a network switch is to receive data frames and forward them to the correct destination device. It learns and stores MAC addresses of connected devices. A switch reduces network collisions by creating separate communication paths. It also improves network performance by sending data only where it is needed instead of broadcasting it to all devices.

How does a switch use MAC address?

A switch uses MAC addresses to identify and forward data to the correct device. When a data frame enters the switch, it reads the destination MAC address in the frame header. The switch checks its MAC address table to find the corresponding port. It then sends the data only through that specific port, ensuring efficient communication.

What is a MAC address table (CAM table)?

A MAC address table, also known as a CAM (Content Addressable Memory) table, is a database maintained by a switch. It stores MAC addresses along with the port numbers to which devices are connected. The switch builds this table automatically by observing incoming traffic. This table helps the switch make fast and accurate forwarding decisions.

Explain the plug-and-play nature of basic network switches.

Basic network switches are plug-and-play devices, meaning they do not require manual configuration. When devices are connected, the switch automatically detects and learns their MAC addresses. It begins forwarding data without needing user setup. This makes installation simple and convenient for small networks.

Hardware Components Required

- PC-PT (x3): End devices acting as network nodes.
- Switch (2960-24TT): A 24-port Cisco Catalyst switch used to interconnect the nodes.
- Copper Straight-Through Cables: The transmission media used for PC-to-Switch connectivity.

Hardware and Connection Setup

- Network ID: 192.168.1.0
- Subnet Mask: 255.255.255.0 Note: PC Naming Convention:

- PC 1: Batch_Roll No._A (e.g., S11_01_A)
- PC 2: Batch_Roll No._B (e.g., S11_01_B)
- PC 3: Batch_Roll No._C (e.g., S11_01_C)

Procedure Step 1: Physical Topology Design

- Open Cisco Packet Tracer and select Network Devices > Switches > 2960. Place it in the center.
- Select End Devices > PC and place three units around the switch.
- Naming: Click on each PC label and rename them following the convention. (e.g. S11_01_A, S11_01_B, and S11_01_C).

Step 2: Hardware Identification

- Click on the Switch and select the Physical tab.
- Zoom in to observe the 24 FastEthernet ports (10/100 Mbps) and 2 GigabitEthernet ports (1000 Mbps).
- Note the status LEDs and the Console port used for administrative access.

Step 3: Physical Cabling

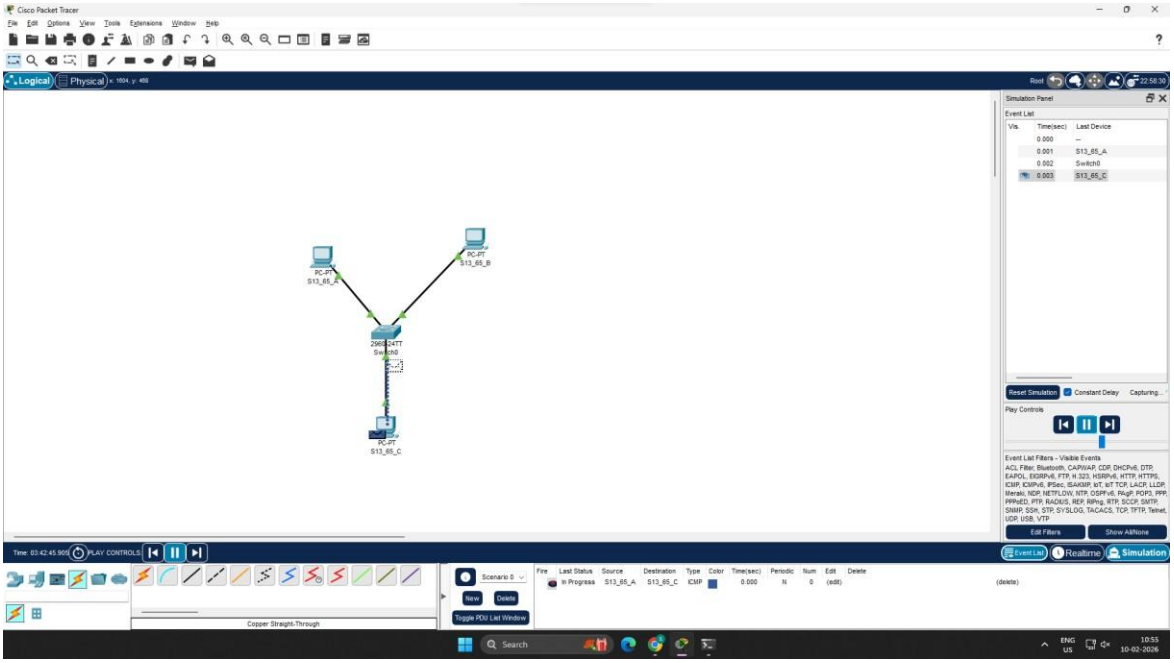
- Select the Copper Straight-Through cable (solid black line).
- Connect S11_01_A (FastEthernet 0) to Switch (FastEthernet 0/1).
- Repeat for the other two PCs using Switch ports 0/2 and 0/3.
- Observation: Notice the link lights are Orange. Click the "Fast Forward Time" button to turn them Green, indicating the Spanning Tree Protocol (STP) has cleared the ports for traffic.

Step 4: Logical Addressing (IP Configuration)

- Click S11_01_A > Desktop > IP Configuration.
- Enter Static IP: 192.168.1.1 and Subnet: 255.255.255.0.
- Configure S11_01_B with IP 192.168.1.2 and S11_01_C with IP 192.168.1.3.

Step 5: Testing and Simulation

- **Real-time:**
  - Open the Command Prompt on S11_01_A and type ping 192.168.1.3. A "Reply from..." message confirms the LAN is functional.
- **Simulation Mode:** o Click the Simulation tab and click Edit Filters to select only ICMP. o Use the Add Simple PDU (Envelope) to click S11_01_A then S11_01_C.
  - Click Play and observe the Switch forwarding the frame only to the destination PC.



Observations

- Media Transition: The Switch ports require time to transition from "Learning" (Orange) to "Forwarding" (Green).
- Micro-segmentation: During simulation, it was observed that the Switch does not broadcast the ICMP packet to all ports; it uses the MAC address table to send it directly to the intended recipient.
- Star Topology Logic: Disconnecting one PC did not affect the communication between the other two, proving the robustness of the Star Topology.

Conclusion

The experiment successfully identifies the hardware features of a Switch and demonstrates the setup of a functional LAN. It concludes that a Switch is the central component of a Star Topology, providing efficient, collision-free data transfer within a single local subnet.